Assessment Item 1

```r
#Set the probability of success
p <- 0.2

#Generate 1000 random variables from the geometric distribution
observe <- 1000
result <- rgeom(n = observe, prob = p)

#Basic statistics
mean_x <- mean(result)
var_x <- var(result)
sd_x <- sd(result)

#Print the results
cat("• Number of trials required to achieve first success:\n")
```

```
## • Number of trials required to achieve first success:
```

```r
cat(sprintf("• Mean (in 2 decimal places): %.2f\n", mean_x))
```

```
## • Mean (in 2 decimal places): 3.91
```

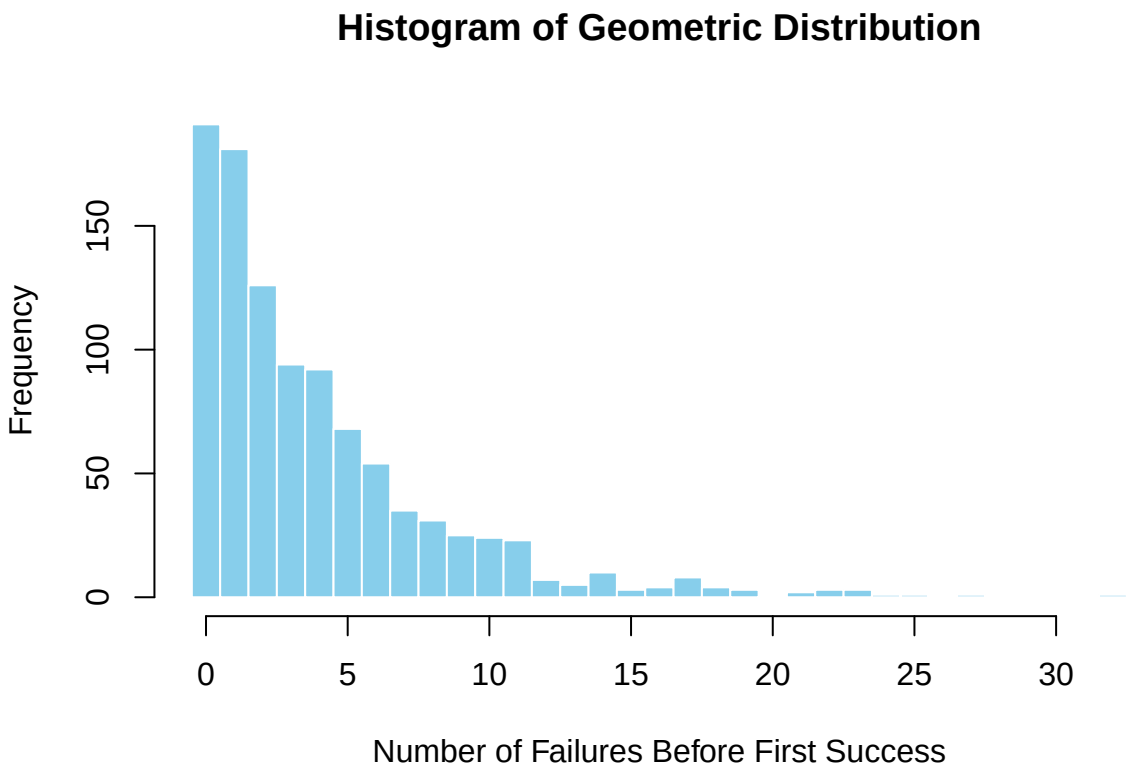
```r
cat(sprintf("• Variance (in 2 decimal places): %.2f\n", var_x))
```

```
## • Variance (in 2 decimal places): 19.39
```

```r
cat(sprintf("• Standard deviation (in 2 decimal places): %.2f\n", sd_x))
```

```
## • Standard deviation (in 2 decimal places): 4.40
```

```r
#Plot the histogram of the results
hist(result,
     breaks = seq(-0.5, max(result) + 0.5, by = 1),
     col = "skyblue",
     border = "white",
     main = "Histogram of Geometric Distribution",
     xlab = "Number of Failures Before First Success",
     ylab = "Frequency")
```



Assessment Item 2 Consider a plant manufacturing IC chips of which 10% are expected to be defective. The chips are packed in boxes for export. Before transportation, a sample is drawn from each box.

```r
# sample of 10 is selected from a box of 40
N1 <- 40
m1 <- N1 * 0.10  # 4 defective chips
n1 <- N1 - m1    # 36 good chips
k1 <- 10         # sample size

# P(X > 1) = 1 - P(X <= 1)
prob_box_40 <- phyper(q = 1, m = m1, n = n1, k = k1, lower.tail = FALSE)

# sample of 10 is selected from a box of 5000.
N2 <- 5000
m2 <- N2 * 0.10  # 500 defective chips
n2 <- N2 - m2    # 4500 good chips
k2 <- 10         # sample size

prob_box_5000 <- phyper(q = 1, m = m2, n = n2, k = k2, lower.tail = FALSE)
```